

Does balanced harvesting still help in a real, calibrated ecosystem?

A mizer reimplementation of Law & Plank (2023), *Fish and Fisheries*

The [main reimplementation](#) tests Law & Plank (2023) the way the paper does it: on ecosystems assembled from scratch inside the model, following the paper’s own Appendix B procedure. On those, the paper’s conclusion reproduces — BHP does not push a single species below 10% of its unfished trajectory, while fixed F and BHP_B do.

That is the right test of the paper. It is not a test of whether the result survives contact with a real ecosystem, and the two can come apart for a reason worth taking seriously: an assembled ecosystem contains rare species *because the assembly procedure produced them*, with life histories drawn at random. A real one contains particular rare species, with particular life histories, at particular abundances that somebody has measured.

The three rules, in the paper’s Eqs (2.7)-(2.9), set the fishing mortality of species i to

$$F_i(t) = F, \quad F_i(t) = c_P P_i(t), \quad F_i(t) = c_{P/B} P_i(t) / B_i(t),$$

where B_i and P_i are the biomass and the somatic production rate over the harvested range — everything above a shared knife-edge entry mass $w_f = 400$ g. All three are size-independent above w_f : the balancing is across species, not across sizes. [The overview](#) writes the rest of the model out in full.

This page runs those three rules on mizer’s North Sea model, `NS_params`, first as it ships and then with real depleted North Sea species added at survey-derived abundances. The short answer is that **the paper’s conclusion holds, but only once the ecosystem actually contains something at risk** — and that the diagnostic usually reached for (“which species is most depleted?”) points the wrong way.

Part 1 — `NS_params` as it ships: BHP buys nothing

`NS_params` is twelve North Sea stocks, calibrated to observations, and it ships fished (effort 0.5–1.0 across its four gears). Setting the three rules up on it is straightforward: a knife edge at $w_f = 400$ g shared by all species, one gear per species so each can carry its own rate, and the rules themselves as a custom `FMort` (see [Implementation](#) below).

Calibrated to equal total yield over 50 years, the outcome is flat:

regime	worst-affected of the twelve
fixed F	0.694
BHP	0.351

regime	worst-affected of the twelve
BH _{P/B}	0.751

BH_P is the *worst* of the three on the conventional most-depleted test. That is not a failure of the rule; it is a property of the ecosystem it was given. Two things drive it:

Nothing in NS_params is at risk. Its `reproduction_level` is 0.99–1.00 for almost every species — recruitment pinned at `R_max`, unable to respond to spawner abundance at all. In a run of these scenarios every recruitment ratio came out at exactly 1.000. A model in which no stock can fail is a model in which a rule designed to stop stocks failing has nothing to do.

The most-depleted species is the most productive one. BH_P sets $F \propto P$, so it concentrates fishing on whatever produces most — here Cod. The “worst-affected species” statistic therefore tracks the species BH_P *intends* to fish hardest, and reports the rule’s design as a defect. Any summary that takes a minimum across species will do this.

So the first finding is methodological rather than ecological: **the standard depletion diagnostic is not a test of balanced harvesting**. It answers a different question from the one the paper asks.

Part 2 — putting real rare species in

The North Sea’s genuinely depleted large species are elasmobranchs, plus Atlantic halibut among teleosts. Four candidates went in: thornback ray (*Raja clavata*), spurdog (*Squalus acanthias*), common skate (*Dipturus batis*) and Atlantic halibut (*Hippoglossus hippoglossus*). All are caught as bycatch by a mixed demersal fishery above 400 g, all grow slowly, and all mature late.

Life-history parameters come from FishBase via `rfishbase`, taking the median across studies:

	w_max	w_mat	k_vb	studies (growth / maturity)
Thornback ray	11.1 kg	2.2 kg	0.124	20 / 17
Spurdog	7.0 kg	1.9 kg	0.080	37 / 17
Common skate	164 kg	27.3 kg	0.057	1 / 2
Atlantic halibut	130 kg	13.4 kg	0.061	8 / 4

For comparison the twelve residents have `k_vb` 0.12–1.00, median 0.30. The rare species are two to five times slower.

The interaction matrix is not a detail

`addSpecies()` defaults every interaction involving a new species to 1, in both directions. In `NS_params` that is far outside the fitted range: the residents’ off-diagonal coefficients have

median 0.24 and maximum 0.565, with 97% below 0.5. The matrix is a spatial-overlap matrix and it is structured — Sole↔Saithe is 0.01, a shallow flatfish against a deep roamer; Cod↔Cod is 0.79.

Taking the default means the new species eat, and are eaten by, all twelve residents more strongly than any fitted pair in the model. It is not a mild assumption, and on this model it decides the answer entirely:

interaction	Thornback ray	Spurdog	Common skate	Halibut
1.0 (the default)	3×10^7	8×10^{16}	2×10^8	1×10^6
0.565 (resident max)	170	2×10^9	460	10
0.239 (resident median)	0.002	36	0.001	0.0002

Those are the reproductive efficiencies `erepro` the species would need. Values above 1 mean more egg biomass than the energy allocated to producing it — not a tuning inconvenience but a physical impossibility, and mizer says so. At the default the species are not merely fragile, they are unrepresentable; two steps down they are comfortable. Everything in between is invisible if you never look.

Rather than pick a constant, each new species inherits the overlap *profile* of the resident it most resembles, which reuses the fitted structure instead of inventing a number:

new species	analogue	rationale
Thornback ray	Gurnard	benthic feeder, medium demersal
Spurdog	Whiting	mobile mid-water piscivore
Common skate	Cod	large demersal predator
Atlantic halibut	Saithe	large, deeper water

The profiles come out sensible without further tuning: halibut barely overlaps sprat (0.02) while skate overlaps nearly everything. At this matrix, unscaled, three of the four species are viable — ray 0.005, skate 0.26, halibut 0.013 — and only spurdog is not.

How rare is rare? Swept-area biomass from the IBTS

Abundance is the other half, and it cannot be guessed: BH_P sets $F \propto P \propto B$, so how rare a species actually is determines directly how much protection the rule appears to give it. `addSpecies()` places new species at whatever is convenient — at most 1/100 of the resource power law — which is an initial condition, not an observation.

Estimates here come from the ICES North Sea IBTS, quarter 1, 2015–2019: 1831 valid hauls, catch converted to mass through FishBase length–weight parameters, divided by the area each net actually fished (tow distance × wing spread, median 19.9 m, matching the GOV trawl’s design), and raised to the 570,000 km² of ICES Subarea 4. Total swept area is 124 km², or 0.022% of the North Sea.

The method validates on a species whose stock size is known independently: **cod comes out at 136,000 t**, against ICES' roughly 100–150 kt for that period.

	swept-area	on NS_params ' scale
Thornback ray	7,890 t	34,800 t
Spurdog	5,020 t	22,200 t
Atlantic halibut	692 t	3,050 t
Common skate	567 t	2,500 t

`NS_params` sits $4.41\times$ above the swept-area estimate for cod, so the second column puts the observations on the model's own scale. Matching to them moves the rare species by **15–40 $\times$** from where `addSpecies()` had parked them: they were far rarer than reality, which would have flattered BH_p considerably.

An earlier attempt scaled them relative to resident species instead and is worth recording as a dead end. It gave estimates spanning four orders of magnitude depending on which resident was chosen — because `NS_params` has Dab at 10.6 kt and Gurnard at 62 kt, while the survey catches $660\times$ more dab than gurnard. The model's relative abundances among its minor species do not match the survey, so anchoring to them inherits that.

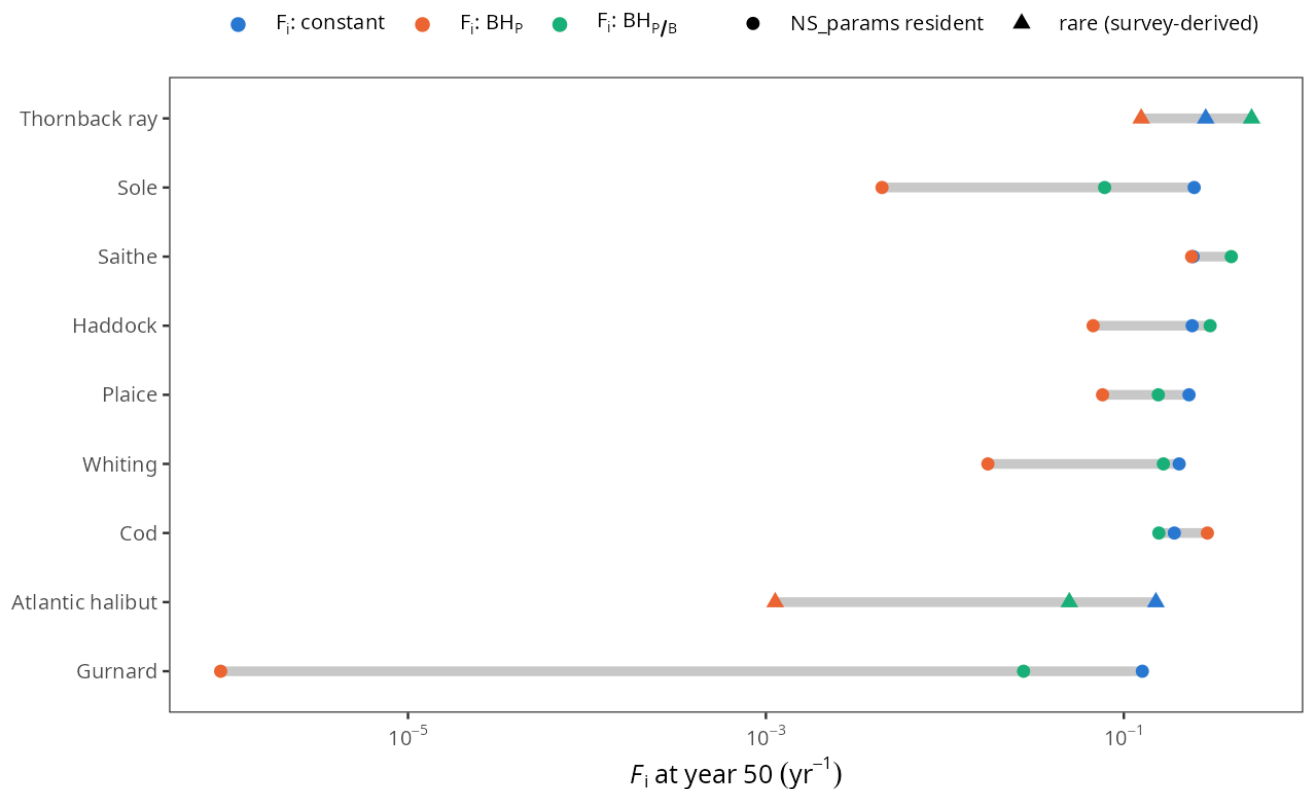
Two species drop out, and one of them is a result

Spurdog cannot be represented at all. It is viviparous, bearing a few ~ 60 g pups after one of the longest gestations of any vertebrate, and mizer's per-species egg size `w_min` is the parameter for that. `addSpecies()` rejects any `w_min` above the model's grid minimum — see [the bug](#) — so spurdog would have to run the entire larval gauntlet from 1 mg, which it cannot.

Common skate needs `erepro` 29 at its observed 2,500 t. The model is saying that the North Sea common skate population cannot be sustained by its own reproduction. That is not obviously wrong. The survey caught 22 individuals in 1831 hauls, every one between 69 and 115 cm, and the species matures at 145 cm — not a single mature animal in five years of sampling. A remnant sustained by immigration, if it is sustained at all, is exactly what fails this test.

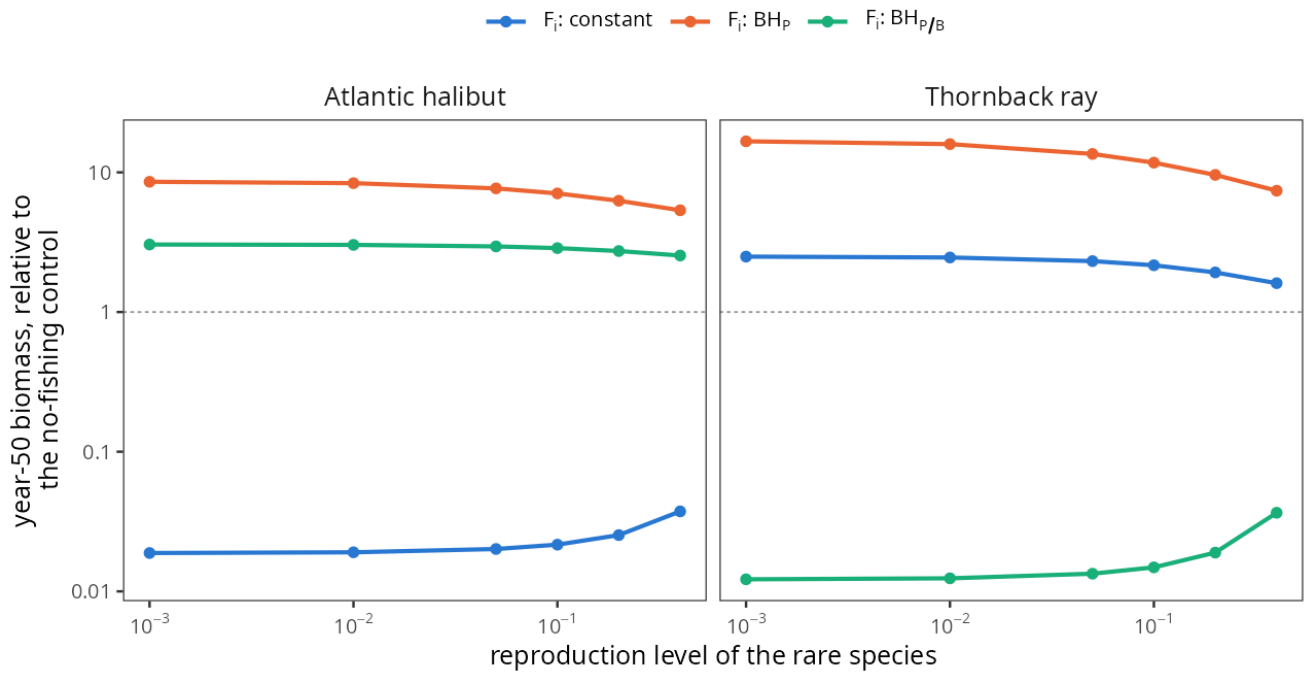
That leaves thornback ray and Atlantic halibut, both viable at their observed abundances (`erepro` 0.645 and 0.449).

The result



How each rule allocates fishing mortality

Each row is a species, each point the fishing mortality one rule assigns it at year 50; all three rules return the same total yield. Constant F is flat by construction. BH_P spans five orders of magnitude, and the two survey-anchored rare species (triangles) sit at its bottom end — halibut at $F = 0.0011 \text{ yr}^{-1}$ against 0.151 under constant F . $BH_{P/B}$ is the interesting one: it is nearly flat, much like constant F , which is the paper's own observation that P/B varies far less across species than P does.



Rare species against the no-fishing control

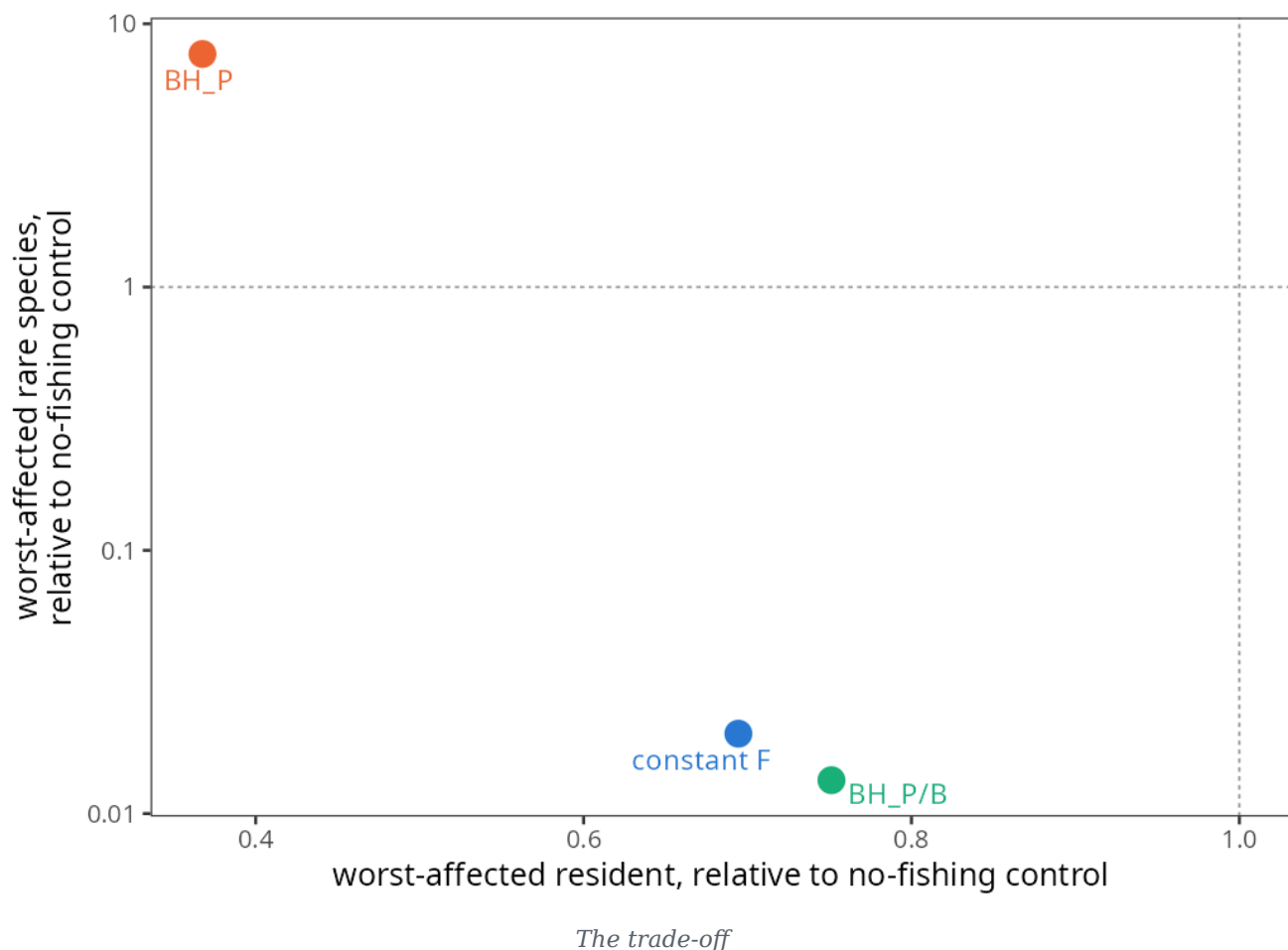
Year-50 biomass against a no-fishing control, so 1.0 means fishing made no difference. Swept over the rare species' reproduction level, because that parameter is free — it leaves the fitted state untouched and then decides how vulnerable a species is.

	constant F	BH_p	$BH_{p/B}$
Atlantic halibut	0.019-0.037	5.4-8.6	2.5-3.0
Thornback ray	1.6-2.5	7.4-16.7	0.012-0.037

Atlantic halibut is destroyed by a size-independent constant F — down to 2-4% of its unfished trajectory — **and protected under BH_p** . That is Law & Plank's claim, on a calibrated ecosystem with an observationally anchored abundance, and it is a factor of roughly 300 between the two rules at identical yield.

The ray goes the other way, and is the more interesting case. **$BH_{p/B}$ destroys it**, assigning $F = 0.518$ — higher than constant F 's 0.287 — because $BH_{p/B}$ targets a high production-to-biomass ratio and the ray, the smaller and faster of the two, has one. Under constant F the ray actually increases: trophic release from fishing down its predators outweighs its own mortality. So which rule is dangerous depends on the species, and $BH_{p/B}$ is dangerous for precisely the kind of species it looks as though it ought to protect.

The direction is robust across the whole feasible reproduction-level range; magnitudes move about twofold.



And this is what BH_P costs. It buys rare-species protection by pressing harder on the productive residents: the worst-affected resident ends at 0.36 of its unfished trajectory, against 0.69 under constant F and 0.75 under BH_{P/B}. The three rules occupy genuinely different corners, and no single number orders them.

Discussion

The paper's conclusion survives the move to a real ecosystem, conditionally. The condition is that the ecosystem has to contain something capable of being lost. `NS_params` as it ships does not: twelve strongly coupled, well-assessed stocks with recruitment pinned at `R_max`. Add two slow-growing, late-maturing species at their measured abundances and the paper's mechanism appears immediately and at full strength.

The most-depleted diagnostic is actively misleading here. It ranks BH_P last in Part 1 and last again in the trade-off figure, while the species it was designed to save goes from 2% to 540% of its unfished trajectory. A rule that redistributes fishing effort towards productive species will always look bad under a statistic that takes a minimum over species without regard to whether the minimum is a healthy stock being worked hard or a rare one being lost. Law & Plank's own framing — counting species pushed below a fraction of their *unfished control* — avoids this, and the main reimplementations use it.

BH_{P/B} is not a mild version of BH_P. It is closer to constant `F` than to BH_P, as the near-flat green series in the first figure shows, and on the thornback ray it is worse than either. This matches the paper’s position that the two readings of “balanced” need to be kept apart, and gives a concrete case where conflating them would cost a species.

Two model-structural findings fell out along the way, both about parameters that are free in the sense of leaving the fitted state untouched while determining the answer:

- *Reproductive compensation.* `setBevertonHolt()` is built to preserve the current steady state, so `reproduction_level` can be set anywhere and the model still fits the data equally well. It then decides how vulnerable a species is. This is why it is swept here rather than chosen.
- *Interaction coefficients.* As above — a factor of 10^{10} in required `erepro` between the default and a plausible value.

Neither is visible if you accept the defaults, and both are the kind of thing that makes a single-configuration result untrustworthy.

Caveats

- **The ray is not stationary.** It drifts to 0.71–0.84 over 50 unfished years despite `isSteady()` returning `TRUE` and its biomass matching the observation exactly. Every regime is therefore reported against the no-fishing control rather than against year 0, which isolates the fishing effect — but the drift itself is not understood. The main reimplementation hits the same problem from the other direction and documents it under [“One correction to how the comparison is framed”](#).
- **Swept-area assumes full retention**, so these biomasses are lower bounds. Catchability differs between rays, flatfish and roundfish, and no correction is applied.
- **The large BH_P increases (5–17×) are trophic release**, not merely the absence of fishing. The rare species expand as the residents are fished down. Read them as “not lost”, not as “five times better off”.
- **Only two species survived to the experiment**, from four candidates, in a model of twelve residents. This is a demonstration that the mechanism operates on real life histories and real abundances, not a stock assessment.
- **NS_params is 4.41× above swept-area for cod**, and disagrees with the survey by more for its minor species. The rare species are matched to observations while the residents are left at their model values, which is internally inconsistent; recalibrating the whole model was out of scope.

A mizer bug found along the way

`steadySingleSpecies()` returns an all-zero spectrum for any species whose `w_min` sits above the model’s minimum grid weight, and `addSpecies()` — which calls it — then fails with an

error that names neither the parameter nor the species:

```
new <- data.frame(species = "Ray", w_max = 9000, w_mat = 2000, w_min = 10)

p <- addSpecies(NS_params, new, steady = FALSE)
sum(initialN(p)["Ray", ] > 0) # 38 - correct
sum(initialN(steadySingleSpecies(p))["Ray", ] > 0) # 0 - emptied

addSpecies(NS_params, new)
#> Error: Candidate steady state holds non-numeric values.
```

With the spectrum all zero, `which.max()` returns index 1 and `addSpecies()` divides by `initial_n[i, idx] == 0`, so the whole array becomes `NaN`. Reported as [sizespectrum/mizer#610](#). This is what blocks spurdog.

Implementation

`project()`'s `effort` argument has to be prescribed in advance, so it cannot express a rule whose rate depends on the evolving state. The lever is `setRateFunction()`: `mizer` computes `e_growth` *before* it calls `FMort` and passes it in, which is exactly what a production-based rule needs.

```
balancedFMort <- function(params, n, n_pp, n_other, t, effort,
                          e_growth, pred_mort, ...) {
  op <- other_params(params)
  sel <- params$selectivity[1, 1, ] # shared knife edge at w_f
  F_i <- switch(op$rule,
    none = op$z * 0,
    fixed = op$const * op$z,
    BHP = op$const * op$z * production(params, n, e_growth),
    BHPB = op$const * op$z * production(params, n, e_growth) /
      biomass_fished(params, n)
  F_i[!is.finite(F_i)] <- 0
  outer(F_i, sel)
}
```

Two things worth knowing if you build on this:

- **Set `knife_edge_size` explicitly.** Its default is `w_mat`, which is per-species — you would silently get a maturity-based fishery rather than a single shared entry size.
- **`c_P` is not portable.** It has dimensions area/mass, so its value belongs to the model's biomass units. Law & Plank's $1 \text{ m}^2 \text{ g}^{-1}$ corresponds to 1.4×10^{-12} here.

Reproducing it

R/ns_model.R	the three rules, as a custom FMort
R/ns_interaction.R	analogy-based interaction matrix for the new species
R/ns_species.R	FishBase life history (needs rfishbase + duckdb)
R/ns_survey.R	IBTS swept-area biomass (needs icesDatras)
R/ns_rare.R	the experiment and the reproduction-level sweep
R/ns_figures.R	the three figures on this page
R/ns_residents.R	Part 1, the twelve residents alone

R/ns_survey.R downloads about 220,000 DATRAS records and caches them to data/datras_ns_ibts_q1.rds , which is deliberately not committed — it is third-party survey data and should be re-downloaded rather than redistributed. Everything else runs from the committed data/ns_*.rds .

Page <https://gustavdelius.github.io/balanced-harvest/north-sea.html> of the site accompanying the mizer reimplementation of Law & Plank (2023). Generated by `make_pdfs.py`.